

AI SD–Assignment–2

(Checkpoint–2,3)



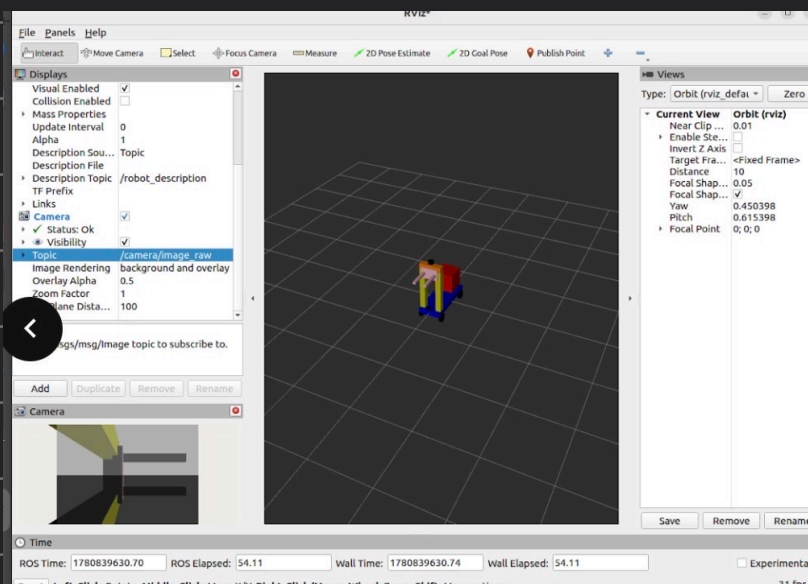
checkpoint - 2

```

305
306 <!--Camera link-->
307 <link name="camera_link">
308   <visual>
309     <geometry>
310       <box size="0.05 0.05 0.05"/>
311     </geometry>
312     <material name="black">
313       <color rgba="0 0 0 1"/>
314     </material>
315     <origin xyz="0 0 0" rpy="0 0 0"/>
316   </visual>
317   <collision>
318     <origin xyz="0 0 0" rpy="0 0 0"/>
319     <geometry>
320       <box size="0.05 0.05 0.05"/>
321     </geometry>
322   </collision>
323   <inertial>
324     <mass value="0.1"/>
325     <origin xyz="0 0 0" rpy="0 0 0"/>
326     <inertia ixx="0.00004" ixy="0" ixz="0" iyy="0.00004" iyz="0" izz="0.00004"/>
327   </inertial>
328 </link>
329
330 <joint name="base_camera_joint" type="fixed">
331   <parent link="base_link"/>
332   <child link="camera_link"/>
333   <origin xyz="0.5 0 0.1" rpy="0 0 0"/>
334 </joint>
335
336 <gazebo reference="camera_link">
337   <sensor type="camera" name="camera">
338     <update_rate>30</update_rate>
339     <camera>
340       <horizontal_fov>1.047</horizontal_fov>
341       <image>
342         <width>640</width>
343         <height>480</height>
344         <format>R8G8B8</format>
345       </image>
346       <clip>
347         <near>0.1</near>
348         <far>100</far>
349       </clip>
350     </camera>
351     <plugin name="gazebo_ros_lidar" filename="libgazebo_ros_laser.so">
352       <topicName>/scan</topicName>
353       <frameName>lidar_link</frameName>
354     </plugin>
355   </sensor>
356 </gazebo>

```

Rviz2



terminal-1

```
command: ros2 run robot_state_publisher robot_state_publisher  
-- ros-args -p robot_description:="$ Cat ~/AISD_ws/src/urdf-  
tutorials/urdf/my_robot.sensors.urdf) "
```

terminal-2:

```
command: rviz2
```

terminal-3:

```
command: ros2 run joint_state_publisher joint_state_publisher
```

Rviz2: ① Fixed Frame = "base-link"

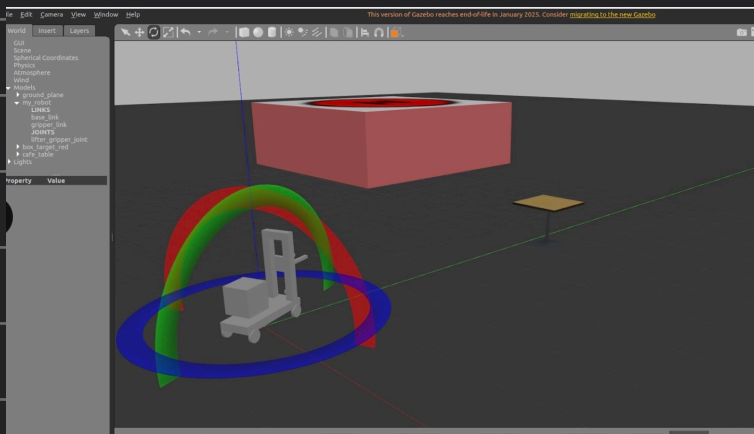
② Add → By Display Type → Robot Model

↓
choose topic as /robot_description

③ Add → By Display Type → Camera

↳ choose topic as /camera/image_raw

Gazebo



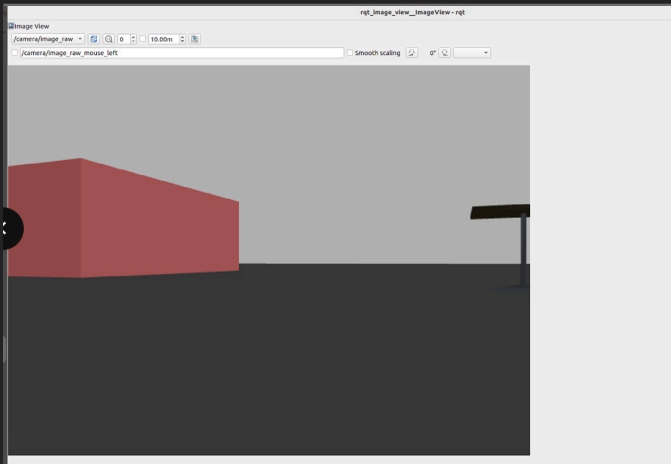
add objects → from insert

terminal-4:

```
Command: ros2 launch gazebo-ros gazebo.launch.py
```

terminal-5:

```
Command: ros2 run gazebo-ros spawn_entity.py -topic  
robot_description -entity my_robot
```



terminal-6:

command: `ros2 run rqt_image_view
rqt_image_view`

command: `ros2 topic hz /camera/image_raw`

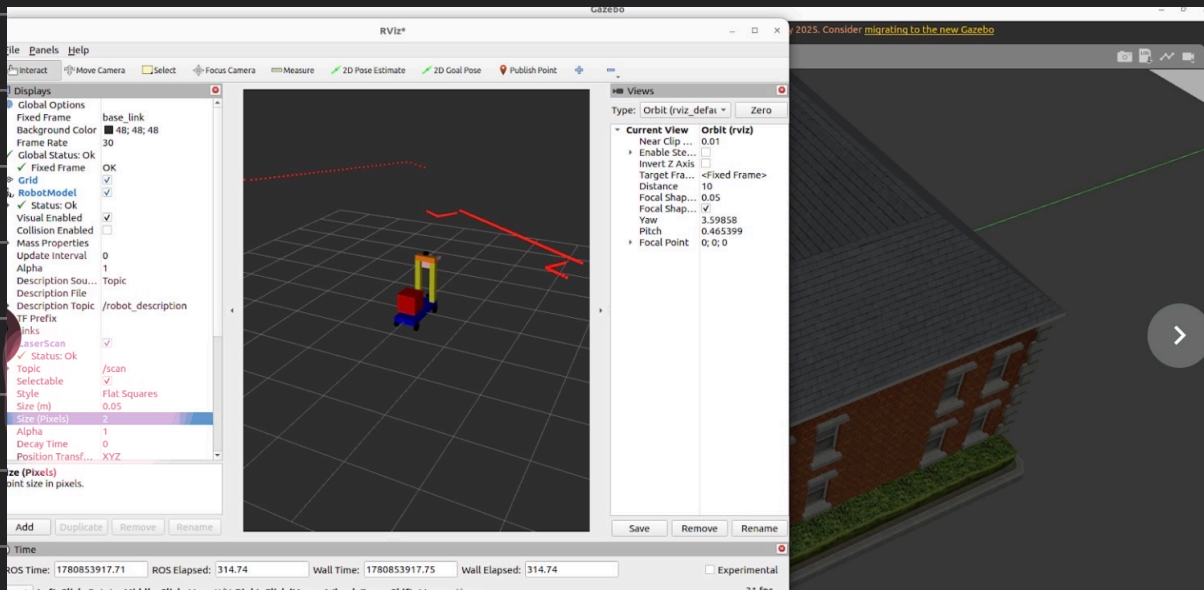
```
357 /
358 <!-- LIDAR -->
359 <link name="lidar link">
360   <visual>
361     <geometry>
362       <cylinder length="0.05" radius="0.05"/>
363     </geometry>
364     <material name="black">
365       <color rgba="0 0 0 1"/>
366     </material>
367     <origin xyz="0 0 0" rpy="0 0 0"/>
368   </visual>
369   <collision>
370     <origin xyz="0 0 0" rpy="0 0 0"/>
371     <geometry>
372       <cylinder length="0.05" radius="0.05"/>
373     </geometry>
374   </collision>
375   <inertial>
376     <mass value="0.1"/>
377     <origin xyz="0 0 0" rpy="0 0 0"/>
378     <inertia ixx="0.00004" ixy="0" ixz="0" iyy="0.00004" iyz="0" izz="0.00004"/>
379   </inertial>
380 </link>
381
382 <joint name="support_lidar_joint" type="fixed">
383   <parent link="support_link"/>
384   <child link="lidar_link"/>
385   <origin xyz="0 0 0.1" rpy="0 0 0"/>
386 </joint>
```

```

388 <gazebo reference="lidar_link">
389 <sensor type="ray" name="lidar">
390 <pose>0 0 0 0 0 0</pose>
391 <visualize>true</visualize>
392 <update_rate>10</update_rate>
393 <ray>
394 <scan>
395 <horizontal>
396 <samples>360</samples>
397 <resolution>1</resolution>
398 <min_angle>-3.14159</min_angle>
399 <max_angle>3.14159</max_angle>
400 </horizontal>
401 </scan>
402 <range>
403 <min>0.1</min>
404 <max>10.0</max>
405 <resolution>0.01</resolution>
406 </range>
407 </ray>
408 <plugin name="gazebo_ros_lidar" filename="libgazebo_ros_ray_sensor.so">
409 <ros>
410 <remapping>~/out:=/scan</remapping>
411 </ros>
412 <output_type>sensor_msgs/LaserScan</output_type>
413 <frame_name>lidar_link</frame_name>
414 </plugin>
415 </sensor>
416 </gazebo>
417
418 </robot>
419

```

Rviz2



terminal-1

command: `ros2 run robot_state_publisher robot_state_publisher`
`-- ros-args -p robot_description:="$ Cat ~/A1SD-ws/src/urdf-tutorials/urdf/my-robot.sensors.urdf )"`

terminal-2:

command: `rviz2`

terminal - 3:

command: `ros2 run joint-state-publisher joint-state-publisher`

Rviz2: ① Fixed Frame = "base-link"

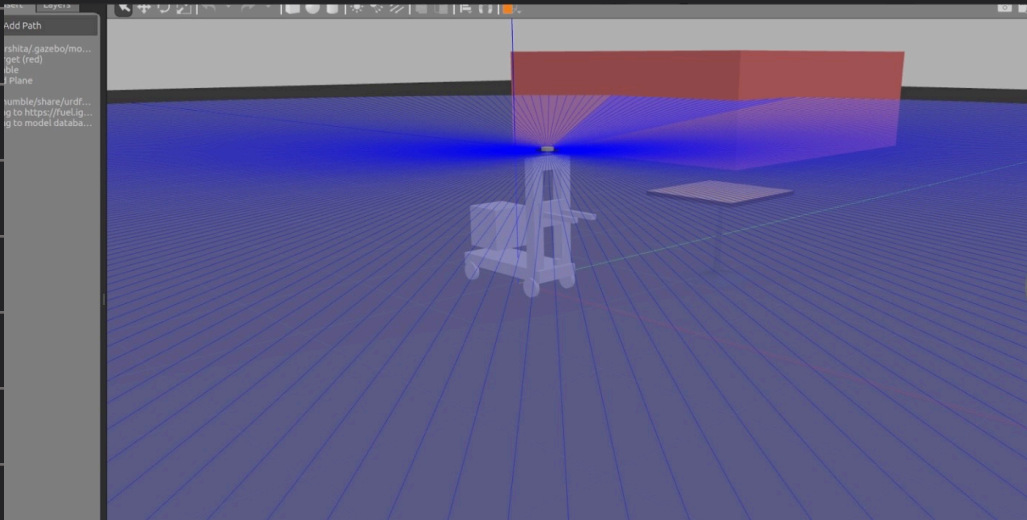
② Add → By Display Type → Robot Model

↓
choose topic as /robot-description

③ Add → By Display Type → Laser Scan

↓
choose topic as /scan

Gazebo



command: `ros2 topic
hz /scan`

terminal - 4:

Command: `ros2 launch gazebo-ros gazebo.launch.py`

terminal - 5:

Command: `ros2 run gazebo-ros spawn-entity.py -topic
robot-description -entity my-robot`

Checkpoint -3

```
421
422 <!--Differential Drive-->
423 <gazebo>
424   <plugin name="differential_drive_controller" filename="libgazebo_ros_diff_drive.so">
425     <ros>
426       <remapping>cmd_vel:=cmd_vel</remapping>
427       <remapping>odom:=odom</remapping>
428     </ros>
429     <update_rate>50</update_rate>
430     <left_joint>base_wheel_1_joint</left_joint>
431     <right_joint>base_wheel_2_joint</right_joint>
432     <wheel_separation>0.4</wheel_separation>
433     <wheel_diameter>0.16</wheel_diameter>
434     <max_wheel_acceleration>1.0</max_wheel_acceleration>
435     <max_wheel_torque>20</max_wheel_torque>
436     <publish_odom>true</publish_odom>
437     <publish_odom_tf>true</publish_odom_tf>
438     <publish_wheel_tf>true</publish_wheel_tf>
439     <odometry_frame>odom</odometry_frame>
440     <robot_base_frame>base_link</robot_base_frame>
441   </plugin>
442 </gazebo>
443
444 </robot>
445
```

terminal-1.

Command: `ros2 run robot_state_publisher robot_state_publisher`
-- `ros-args -p robot_description:="$(cat ~/A1SD-ws/src/vrdp-tutorials/vrdp/my-robot-in-motion.vrdf)"`

terminal-4:

Command: `ros2 launch gazebo-ros gazebo.launch.py`

terminal-5:

Command: `ros2 run gazebo-ros spawn-entity.py -topic robot_description -entity my-robot`

